

The background of the slide features abstract, overlapping green geometric shapes, primarily triangles and polygons, in various shades of green, creating a modern and dynamic visual effect. A solid dark gray horizontal band spans the width of the slide, serving as a backdrop for the title text.

Memory Design

Introduction

- Aim:

To design a desired memory configuration with available memory chips.

Steps:

1. To find the number of chips required.
2. To find the number of address lines required.
3. Memory Address Map.
4. Memory Design.

Memory Design

- Available Memory chip Size $M_{N, W} : N \times W$
- Required memory size: $N^1 \times W^1$, Where $N^1 \geq N$ and $W^1 \geq W$
- Required number of $M_{N, W}$ chips: $p \times q$, Where $p = \lceil N^1 / N \rceil$ and $q = \lceil W^1 / W \rceil$

Memory design

There are 3 types of organizations of $N^1 \times W^1$ that can be formed using $N \times W$

- $N^1 = N$ and $W^1 > W \Rightarrow$ increasing the word size of the chip
- $N^1 > N$ and $W^1 = W \Rightarrow$ increasing the number of words in the memory
- $N^1 > N$ and $W^1 > W \Rightarrow$ increasing both the number of words and number of bits in each word.

Memory design – Increasing the word size

- Problem - 1

- Design 128×16 - bit RAM using 128×4 - bit RAM
- Solution: $p = 128 / 128 = 1$; $q = 16 / 4 = 4$
- Therefore, $p \times q = 1 \times 4 = 4$ memory chips of size 128×4 are required to
- construct 128×16 bit RAM

S.No	Memory Type	$N \times W$	$N^1 \times W^1$	p	q	$p * q$	x	y	z	Total
1	RAM	128×4	128×16	1	4	4	7	0	0	7

x – number of address lines

y ($p = 2^y$)

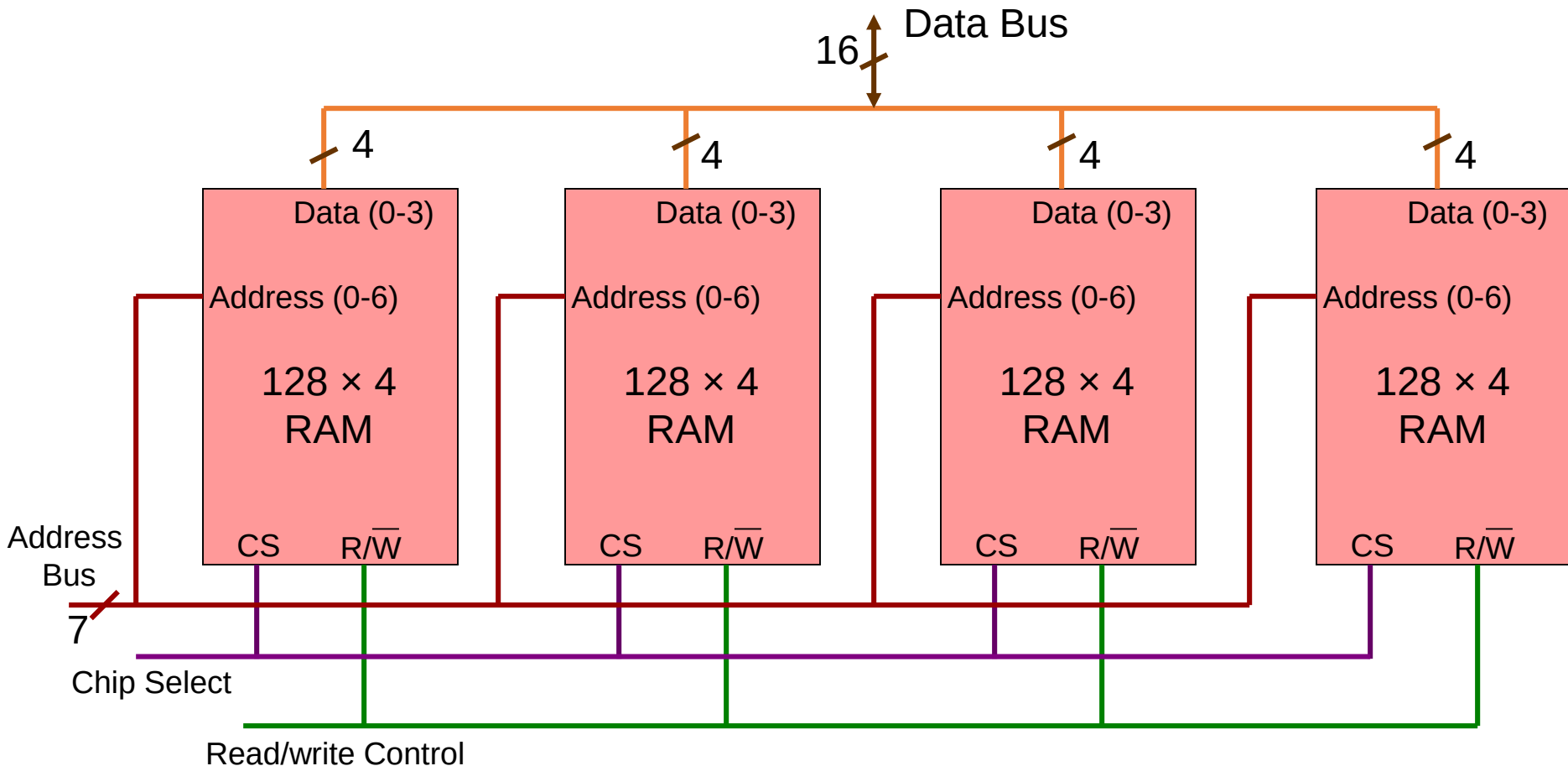
z – to select the type of memory

Memory Address Map

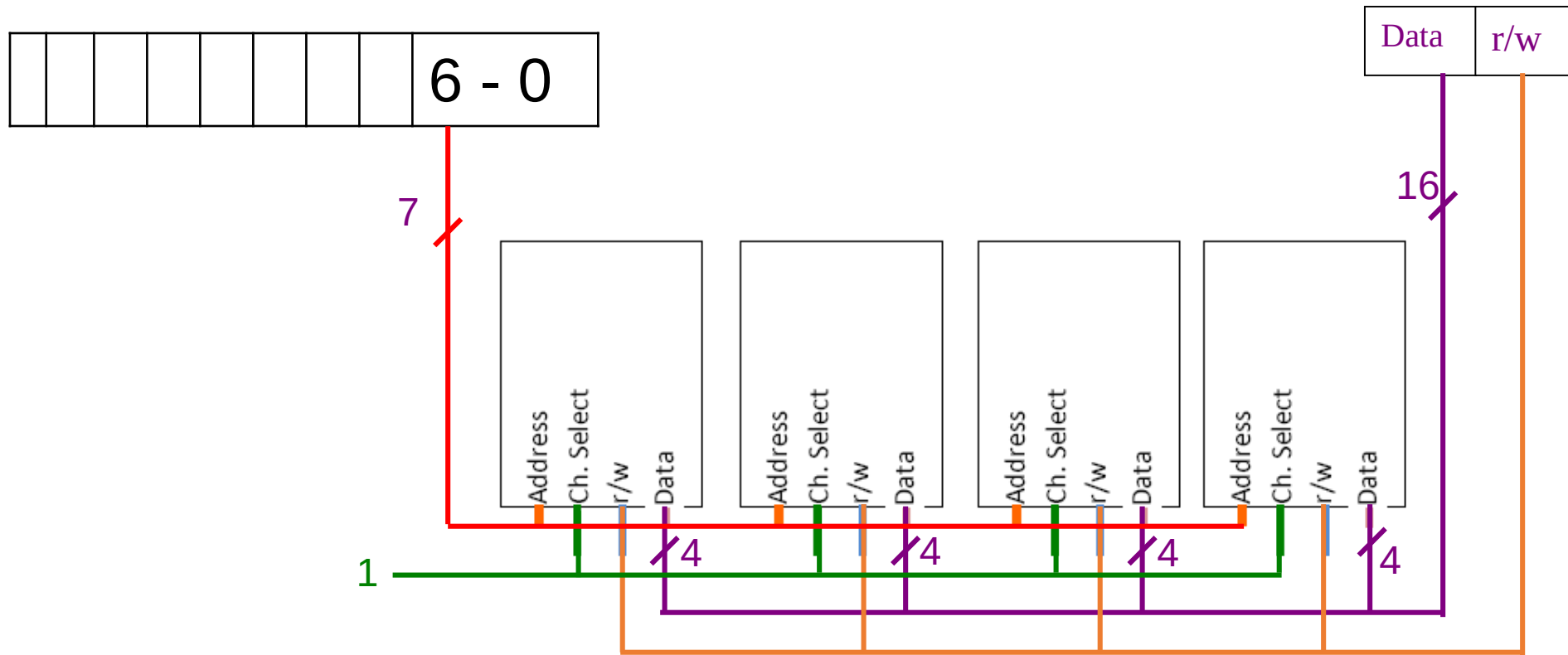
Component	Hexadecimal address		Address Bus															
	From	To	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RAM 1.1	0000	007F										0	0	0	0	0	0	0
RAM 1.2	0000	007F										X	X	X	X	X	X	X
RAM 1.3	0000	007F										X	X	X	X	X	X	X
RAM 1.4	0000	007F										X	X	X	X	X	X	X

Substitute 0 in place of x to get 'From' address and 1 to get 'To' address

Memory design – Increasing the word size



Memory Design



Memory Design – Increasing the number of words

- Problem - 1

- Design 1024×8 - bit RAM using 256×8 - bit RAM
- Solution: $p = 1024 / 256 = 4$; $q = 8 / 8 = 1$
- Therefore, $p \times q = 4 \times 1 = 4$ memory chips of size 256×8 are required to construct 1024×8 bit RAM

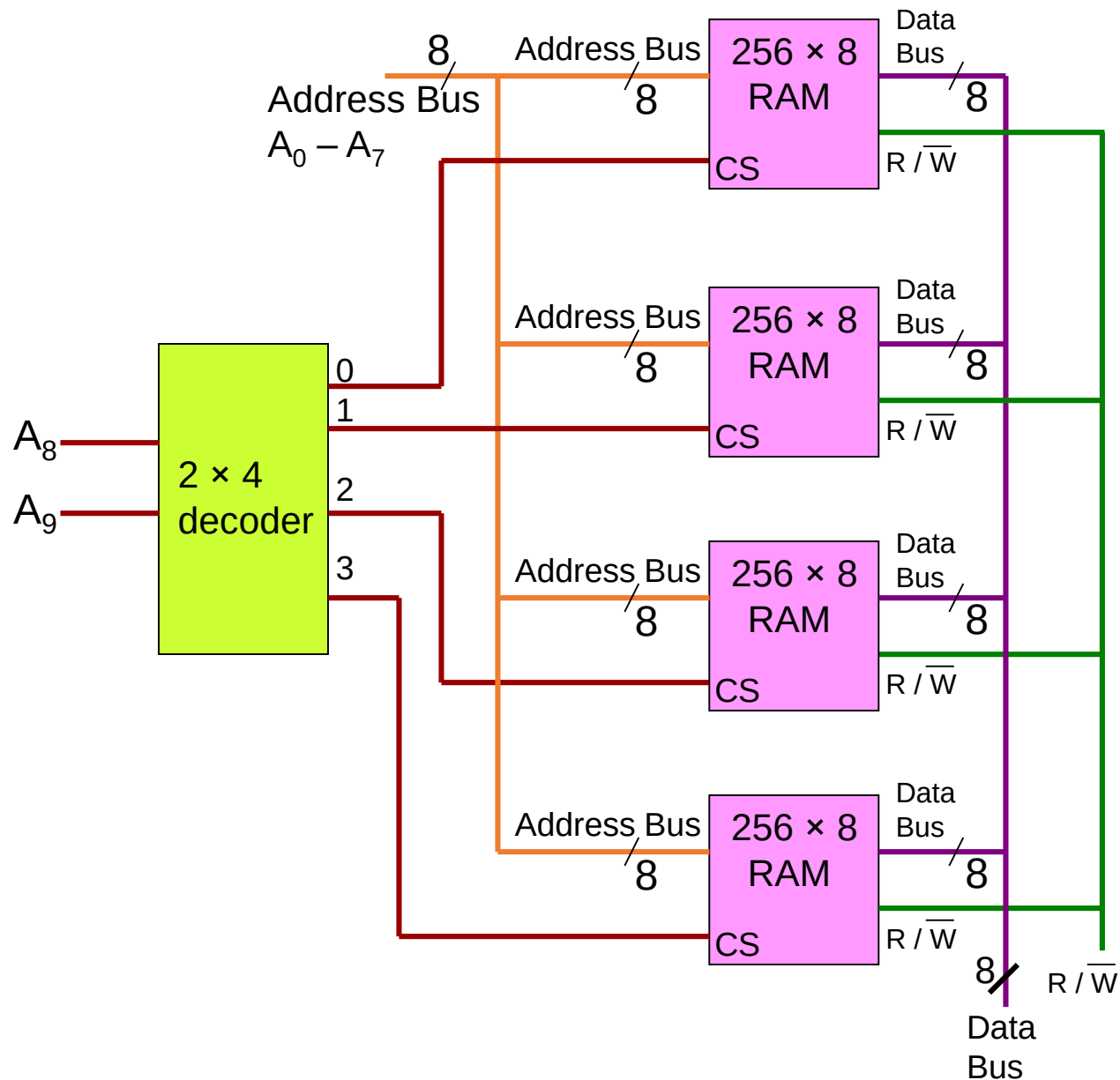
[illegible]

Memory Address Map

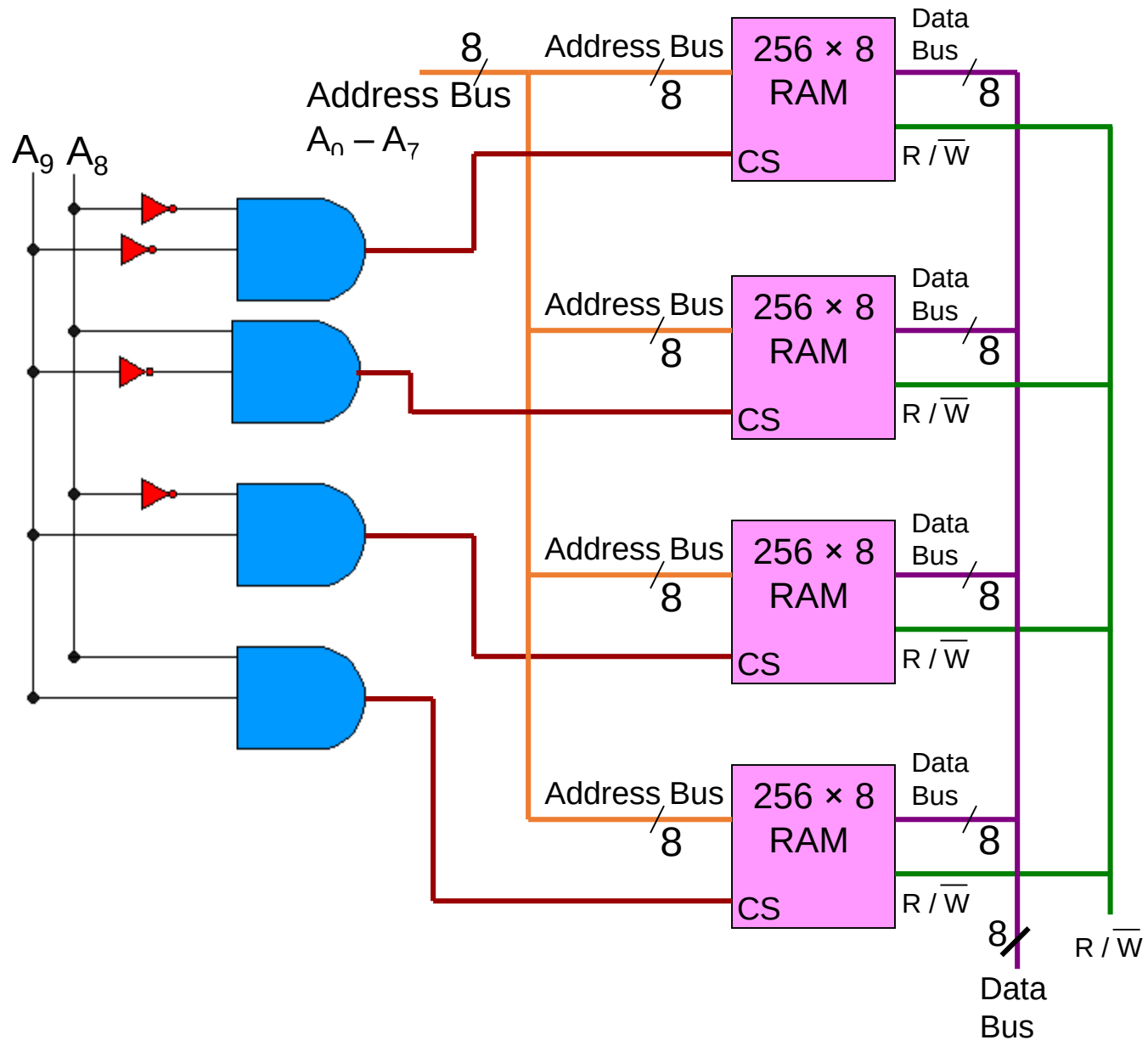
Component	Hexadecimal address		Address Bus															
	From	To	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RAM 1	0000	00FF							0	0	x	x	x	x	x	x	x	x
RAM 2	0100	01FF							0	1	x	x	x	x	x	x	x	x
RAM 3	0200	02FF							1	0	x	x	x	x	x	x	x	x
RAM 4	0300	03FF							1	1	x	x	x	x	x	x	x	x

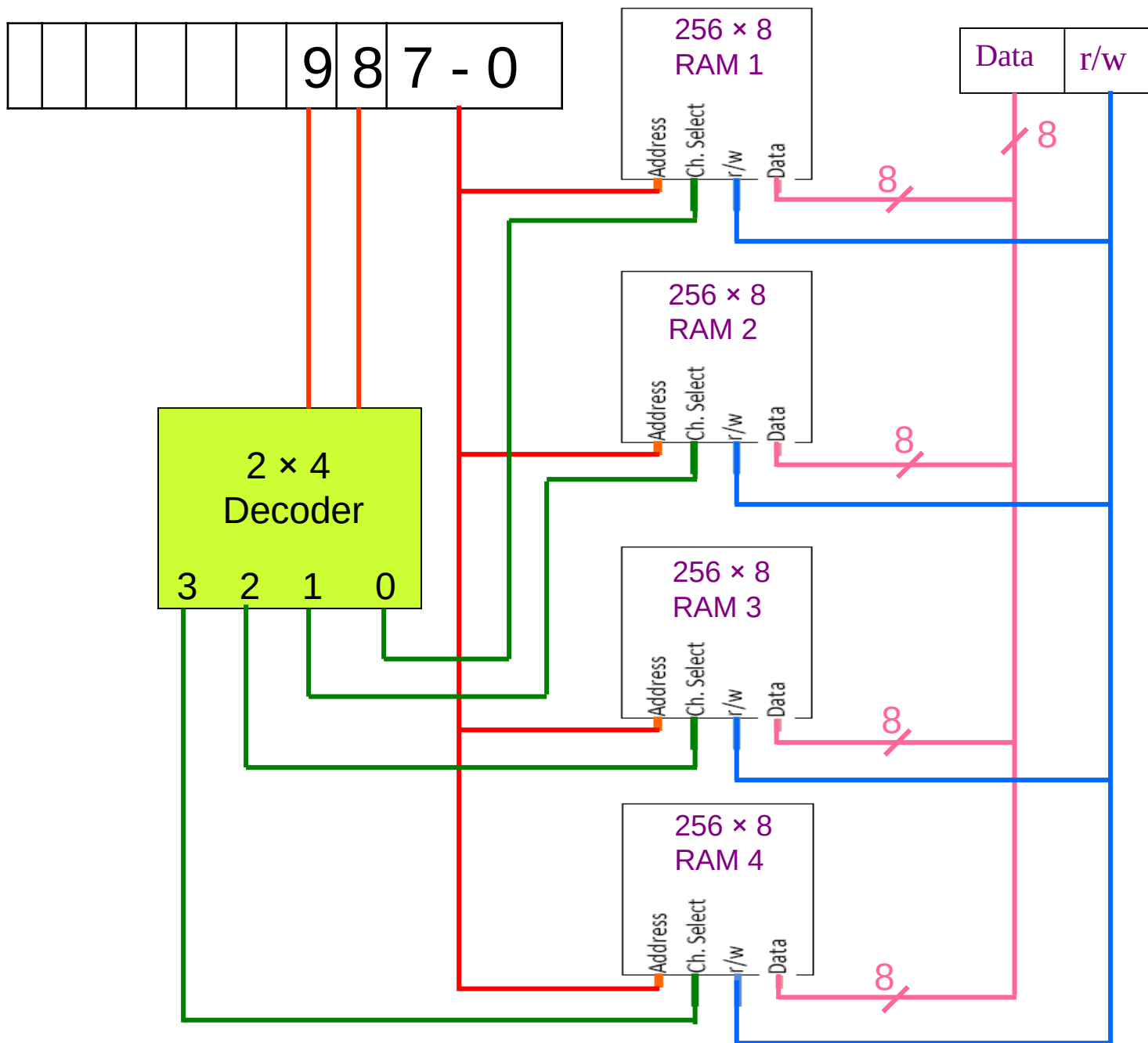
Substitute 0 in place of x to get 'From' address and 1 to get 'To' address

Memory Design – Increasing the number of words



Design with gates





Memory Design

- Problem - 2

- Design 256×16 – bit RAM using 128×8 – bit RAM chips

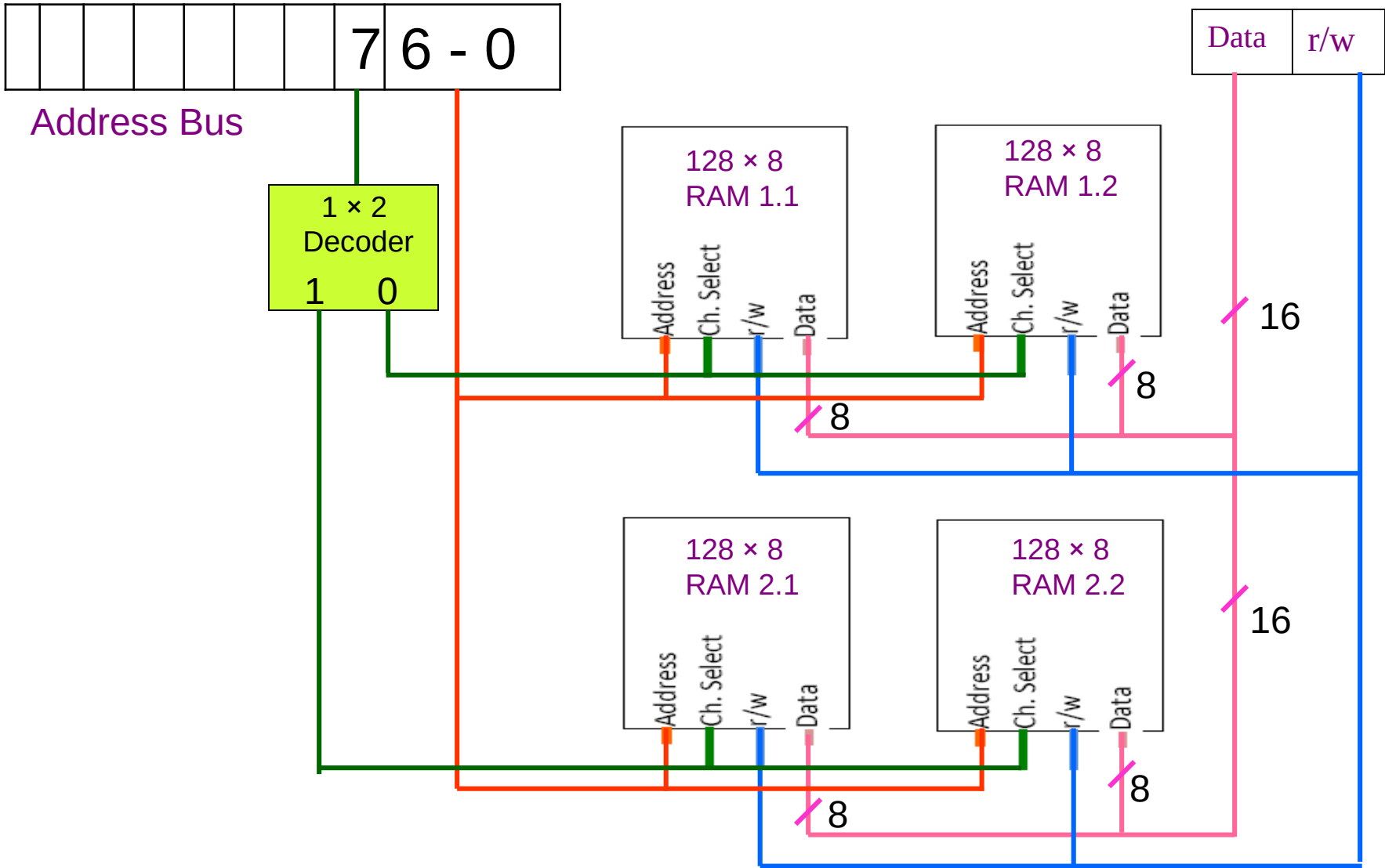
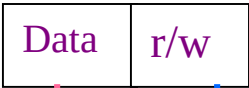
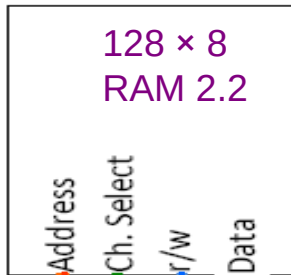
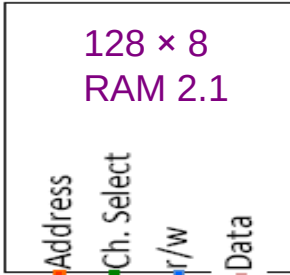
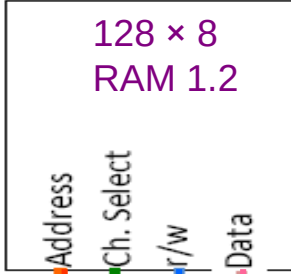
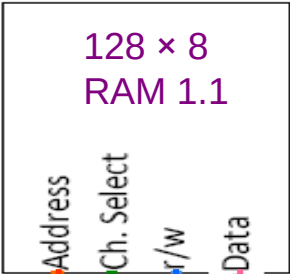
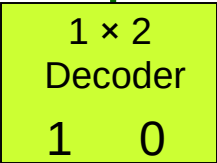
[illegible]

Memory Address Map

Component	Hexadecimal address		Address Bus															
	From	To	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RAM 1.1	0000	007F									0	x	x	x	x	x	x	x
RAM 1.2	0000	007F									0	x	x	x	x	x	x	x
RAM 2.1	0080	00FF									1	x	x	x	x	x	x	x
RAM 2.2	0080	00FF									1	x	x	x	x	x	x	x



Address Bus



Memory Design

- Problem - 3

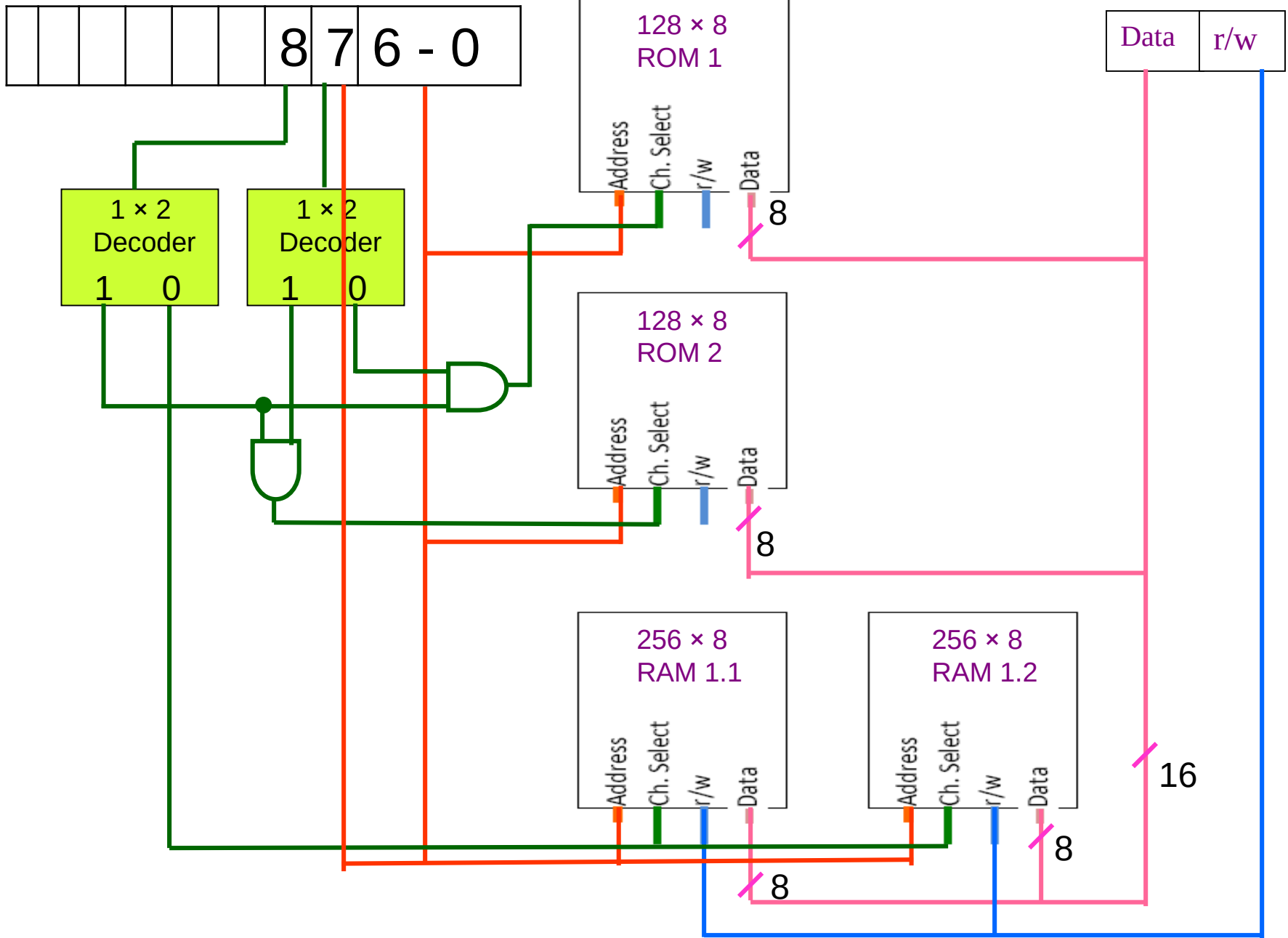
- Design 256×16 – bit RAM using 256×8 – bit RAM chips and 256×8 – bit ROM using 128×8 – bit ROM chips.

[illegible]

Memory Address Map

Component	Hexadecimal address		Address Bus															
	From	To	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RAM 1.1	0000	00FF								0	x	x	x	x	x	x	x	x
RAM 1.2	0000	00FF								0	x	x	x	x	x	x	x	x
ROM 1	0100	017F								1	0	x	x	x	x	x	x	x
ROM 2	0180	01FF								1	1	x	x	x	x	x	x	x

Address Bus



Memory design

- Problem – 4
- A computer employs RAM chips of 128×8 and ROM chips of 512×8 . The computer system needs 256 bytes of RAM, 1024×16 of ROM, and two interface units with 256 registers each. Design the chip layout for the above description.

Requirements

S.NO	Memory	N x W	N ¹ x W ¹	P	q	p * q	x	y	z	Total
1	RAM	128 × 8	256 × 8	2	1	2	7	1	2	10
2	ROM	512 × 8	1024 × 16	2	2	4	9	1	2	12
3	Interface	256		2	1	2	8	1	2	11
4										

q is 1 always for interfaces.

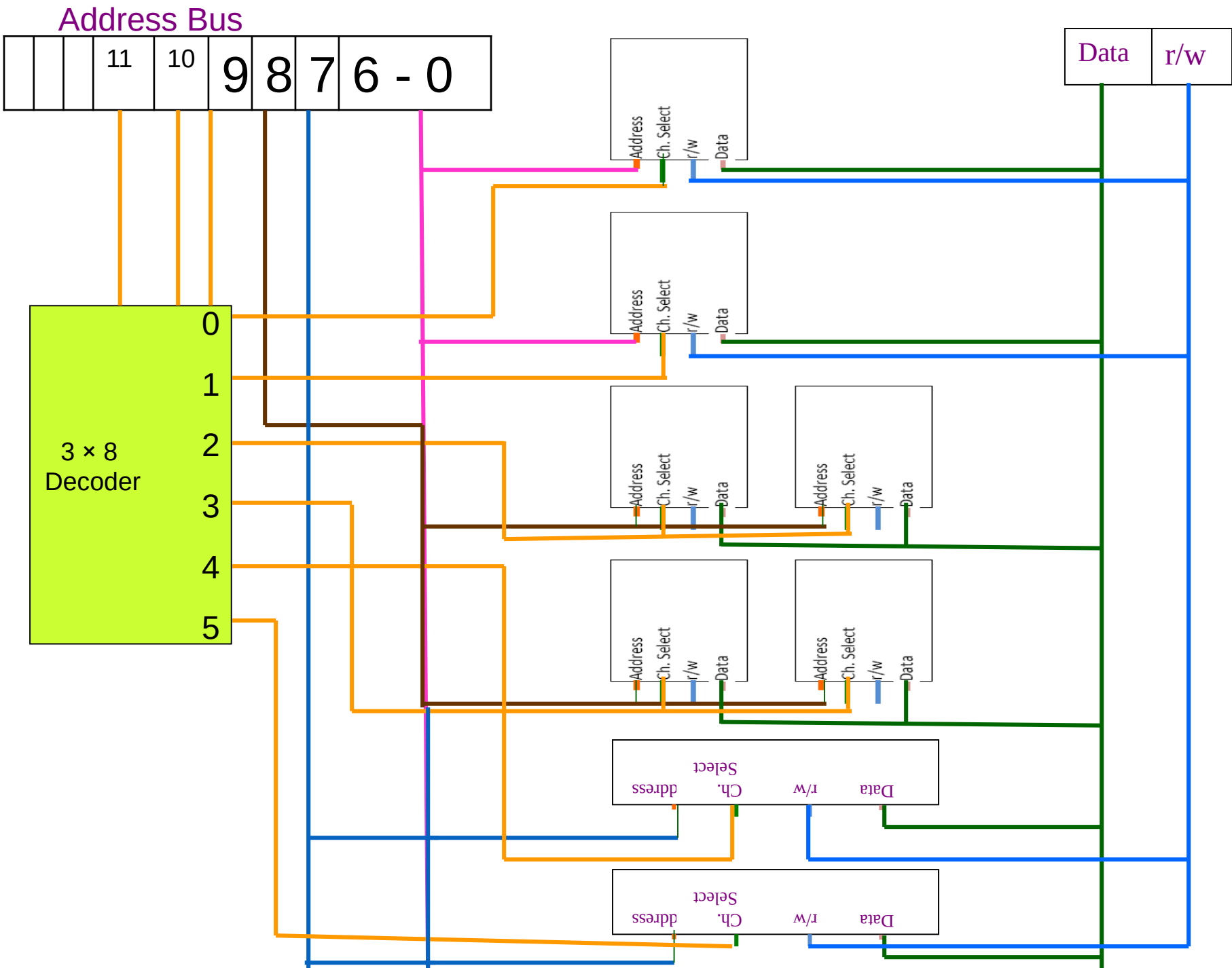
Number of registers = 2^x

P = number of interfaces

Number of data lines = size of registers

Memory Address Map

Component	Hexadecimal Address		Address Bus												
	From	To	15 - 12	11	10	9	8	7	6	5	4	3	2	1	0
RAM1	0000	007F		0	0	0			x	x	x	x	x	x	x
RAM2	0200	027F		0	0	1			x	x	x	x	x	x	x
ROM1.1	0400	05FF		0	1	0	x	X	x	x	x	x	x	x	x
ROM1.2	0400	05FF		0	1	0	x	X	x	x	x	x	x	x	x
ROM2.1	0600	07FF		0	1	1	x	X	x	x	x	x	x	x	x
ROM2.2	0600	07FF		0	1	1	x	X	x	x	x	x	x	x	x
Interface 1	0800	08FF		1	0	0		X	x	x	x	x	x	x	x
Interface 2	0A00	0AFF		1	0	1		x	x	x	x	x	x	x	x



Assignment Problems

- A computer employs RAM chips of 1024×8 and ROM chips of 512bytes. Design the memory system which needs $2K \times 32$ of RAM, $1K \times 8$ of ROM and an interface unit with 128 registers.

Assignment Problems

- a. How many 128 x 8 RAM chips are needed to provide a memory capacity of 2048 bytes?
- b. How many lines of the address bus must be used to access 2048 bytes of memory? How many of these lines will be common to all chips?
- c. How many lines must be decoded for chip select? Specify the size of the decoder

A computer uses RAM chips of 1024 x 1 capacity.

- How many chips are needed, and how should their address lines be connected to provide a memory capacity of 1024 bytes?
- How many chips are needed to provide a memory capacity of 16K bytes. Explain in words how the chips are to be connected to the address bus.

Assignment Problems

- A computer employs RAM chips of 256×8 and ROM chips of 1024×8 . The computer system needs 2K bytes of RAM, 4K bytes of ROM, and four interface units, each with four registers. A memory-mapped I/O configuration is used. Assign 00 for RAM, 01 for ROM, and 10 for interface registers.
- a. How many RAM and ROM chips are needed?
- b. Draw a memory-address map for the system.
- c. Give the address range in hexadecimal for RAM, ROM, and interface .
- d. Show the chip layout for the above design

References

- **Computer System Architecture – M. Morris Mano.**
- **Carl Hamacher, Zvonko Vranesic, Safwat Zaky, Computer organization, Mc Graw Hill, Fifth edition ,Reprint 2011.**